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<https://doi.org/10.1063/1.1747673>



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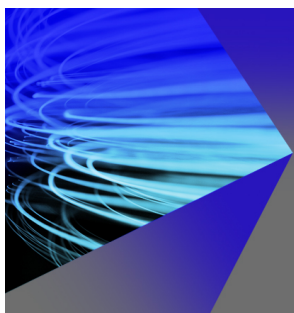
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A Viscosity Equation for Gas Mixtures

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(Received October 24, 1949)

By application of the kinetic theory, with several simplifying assumptions, the previous equation of Buddenberg and the author has been modified to give a general equation for viscosity as a function of molecular weights and viscosities of the pure components of the mixture. Agreement of the equation with experimental data is demonstrated for a number of highly irregular binary gas systems and mixtures of three to seven components.

RECENTLY, Curtiss and Hirschfelder^{1,2} have developed relations for calculation of gas mixture viscosities, based on quantitative evaluation of intermolecular forces and the use of collision integrals, which reproduce the experimental data for non-polar gases with high precision. In the present paper are presented alternate, although less rigorous, equations which may prove useful from the standpoint of simplicity and convenience in numerical use, and which are believed sufficiently accurate for many purposes.

Buddenberg and Wilke³ have shown that the Sutherland type equation⁴ may be expressed in the general form for binary mixtures given by Eq. (1).

$$\mu_m = \frac{\mu_1}{1 + (1.385\mu_1x_2/D_{12}\rho_1x_1)} + \frac{\mu_2}{1 + (1.385\mu_2x_1/D_{12}\rho_2x_2)} \dots, \quad (1)$$

where the dimensionless constant 1.385 is an average value determined empirically by examination of a large body of data.

Equation (1) was further extended to include multi-component systems in the general form:

$$\mu_m = \sum_{i=1}^n \frac{\mu_i}{1 + \left[\frac{1.385\mu_i}{x_i\rho_i} \sum_{j=1}^n \frac{x_j}{D_{ij}} \right]} \quad (2)$$

A major source of uncertainty in the use of these equations lies in the need for the diffusion coefficients for the various gas pairs involved, so that it is often necessary to estimate the diffusion coefficients from empirical correlations.

Equation (1) may be written in a more general form:

$$\mu_m = \frac{\mu_1}{1 + (\beta\mu_1x_2/D_{12}\rho_1x_1)} + \frac{\mu_2}{1 + (\beta\mu_2x_1/D_{21}\rho_2x_2)} \dots, \quad (3)$$

¹ C. F. Curtiss and J. O. Hirschfelder, "The kinetic theory of multicomponent systems of gases," University of Wisconsin, CF-727 (July, 1947).

² Hirschfelder, Bird, and Spotz, Chem. Rev. 44, 205 (1949).

³ J. W. Buddenberg and C. R. Wilke, Ind. Eng. Chem. 41, 1349 (1949).

⁴ W. Sutherland, Phil. Mag. 40, 421 (1895).

where β is a constant for the gas pair under consideration. Considering Eq. (3) for the special case of a mixture of component 1 in itself, say for 50 percent of 1 in 1, and solving for β gives the result:

$$\beta = \beta_1 = (D_{11}\rho_1)/\mu_1. \quad (4)$$

Similarly, considering component 2 alone:

$$\beta = \beta_2 = (D_{22}\rho_2)/\mu_2. \quad (5)$$

According to the kinetic theory⁵ the group $D\rho/\mu$ for self-diffusion ranges in numerical value from 1.2 for hard spheres to 1.543 for inverse-fifth-power repulsion, which supports the average value of 1.385 obtained for the constant in Eq. (1). It is probable, therefore, that β is not equal to a single constant, but rather that it may have various values in the vicinity of 1.385, depending upon the gases under consideration.

The diffusion coefficient in a binary mixture is given by the expression:⁵

$$D_{12} = (1 + \lambda_{12}) \left(\frac{\pi}{2} \right)^{\frac{1}{2}} \frac{1}{nS_{12}} \left(\frac{m_1 + m_2}{m_1 m_2} kT \right)^{\frac{1}{2}} \dots \quad (6)$$

where for hard spherical molecules S_{12} is given by the relation

$$S_{12} = (\pi/4)(\sigma_1 + \sigma_2)^2 \dots \quad (7)$$

Writing (6) and (7) for self-diffusion yields expressions for σ_1 and σ_2 as functions of the coefficients of self-diffusion. Substituting these values of σ_1 and σ_2 in (6) for the binary case, and assuming $(1 + \lambda_{11}) \approx (1 + \lambda_{22}) \approx (1 + \lambda_{12})$, gives an expression for the binary diffusion coefficient in terms of the coefficients of self-diffusion:

$$D_{12} = \frac{(4/\sqrt{2})(m_1 + m_2/m_1 m_2)^{\frac{1}{2}}}{[(1/D_{11}^2 m_1)^{\frac{1}{2}} + (1/D_{22}^2 m_2)^{\frac{1}{2}}]^2} \dots \quad (8)$$

Equations (4) and (5) may be combined with (8) to give a further expression for the diffusion coefficient:

$$D_{12} = \frac{[4/\sqrt{2}(1/M_1 + 1/M_2)^{\frac{1}{2}}]}{[(\rho_1^2/\beta_1^2 \mu_1^2 M_1)^{\frac{1}{2}} + (\rho_2^2/\beta_2^2 \mu_2^2 M_2)^{\frac{1}{2}}]^2} \dots \quad (9)$$

Assuming $\beta = \beta_1 = \beta_2$, Eqs. (3) and (9) may be com-

⁵ E. H. Kennard, *Kinetic Theory of Gases* (McGraw-Hill Book Company, Inc., New York, 1938), p. 195.

TABLE I. Properties of gases used in Fig. 1 and Table II.*

Gas pair	$t^\circ\text{C}$	$\mu_1 \times 10^6$	$\mu_2 \times 10^6$	ϕ_{12}	ϕ_{21}	Reference for viscosity data
He-A	20	197.3	221.1	2.417	0.2716	a, b
N ₂ -O ₂	27	178.1	205.7	0.994	1.014	c
C ₂ H ₆ -C ₃ H ₈	250	152.6	136.3	1.276	0.779	d
H ₂ -CCl ₂ F ₂	25	88.4	124.0	3.943	0.0920	e
H ₂ -C ₃ H ₈	26.9	89.1	81.7	3.674	0.1539	f
H ₂ -A	20	87.5	221.1	1.870	0.2384	a
Ne-A	200	422.0	320.8	1.606	0.6166	a
Ne-He	200	422.0	267.2	0.4862	1.552	a
He-A	200	267.2	320.8	2.318	0.2789	a
N ₂ -CO ₂	25	176.8	147.8	1.368	0.7281	e
H ₂ -CO ₂	25	88.4	147.8	2.470	0.1977	e
H ₂ -N ₂	25	88.4	176.8	1.905	0.2750	e
Ne-CO ₂	25	313.3	147.8	2.250	0.5011	e
Ne-CCl ₂ F ₂	25	313.3	124.0	3.987	0.2628	e
Ne-N ₂	25	313.3	176.8	1.612	0.6552	e
CO ₂ -CCl ₂ F ₂	25	147.8	124.0	1.751	0.5351	e
N ₂ -CCl ₂ F ₂	25	176.8	124.0	2.358	0.3830	e

* All data at atmospheric pressure.

a. M. Trautz and H. E. Benkele, Ann. d. Physik 5, 561 (1930).

b. M. Trautz and K. F. Kipphan, Ann. d. Physik 2, 743 (1929).

c. M. Trautz and A. Melster, Ann. d. Physik 7, 409 (1930).

d. M. Trautz and K. G. Sorg, Ann. d. Physik 10, 81 (1931).

e. J. W. Buddenberg, M.S. thesis, University of California (1948). Data determined relative to N₂.

f. M. Trautz and F. Kurz, Ann. d. Physik 9, 992 (1931).

bined* to eliminate the diffusion coefficient from the viscosity equation:

$$\mu_m = \frac{\mu_1}{1 + \frac{x_2/x_1 \{1 + [(\mu_1/\mu_2)(\rho_2/\rho_1)]^{1/2} (M_1/M_2)^{1/2}\}^2}{(4/\sqrt{2})[1 + (M_1/M_2)]^{1/2}}} + \frac{\mu_2}{1 + \frac{x_1/x_2 \{1 + [(\mu_2/\mu_1)(\rho_1/\rho_2)]^{1/2} (M_2/M_1)^{1/2}\}^2}{(4/\sqrt{2})[1 + (M_2/M_1)]^{1/2}}} \dots (10)$$

For ideal gas behavior, Eq. (10) may be reduced to the relatively simple expression:

$$\mu_m = \frac{\mu_1}{1 + \frac{(x_2/x_1)[1 + (\mu_1/\mu_2)^{1/2} (M_2/M_1)^{1/2}]^2}{(4/\sqrt{2})[1 + (M_1/M_2)]^{1/2}}} + \frac{\mu_2}{1 + \frac{(x_1/x_2)[1 + (\mu_2/\mu_1)^{1/2} (M_1/M_2)^{1/2}]^2}{(4/\sqrt{2})[1 + (M_2/M_1)]^{1/2}}} \dots (11)$$

Equation (11) constitutes the final recommended relation for binary mixtures. This relation is relatively convenient to use, requiring only the molecular weights and viscosities of the pure components for the prediction of the entire viscosity concentration curve. It should be noted that the equation contains no empirical

* Values of β_1 and β_2 need not necessarily be viewed as equal to a single value of β in combining (3) and (9), but rather that $\beta = \beta_1$, in calculating D_{12} , and $\beta = \beta_2$ in calculating D_{21} . Thus it is possible that errors due to deviations between β_1 and β_2 may fortuitously cancel in each of the partial viscosity terms in (3).

TABLE II. Calculated and experimental data for multicomponent mixtures.

$t, ^\circ\text{C}$	Volume percent of components							Viscosity $\times 10^6$, g/cm-sec.	
	Ne	A	He	H ₂	CO ₂	N ₂	CCl ₂ F ₂	μ_{calc}	μ_{exp}
200	31.93	32.13	35.94					360.0	357.4 ^a
25	33.33			33.33	33.33			190.2	185.7 ^b
25				33.33		33.33	33.33	145.7	146.9 ^b
25					33.33	33.33	33.33	144.6	145.8 ^b
25	25.00			25.00	25.00		25.00	159.3	168.1 ^b
25				25.00	25.00	25.00	25.00	146.7	147.1 ^b

a. See Table I, reference b.

b. See Table I, reference e.

numerical constants, the constant $4/\sqrt{2}$ coming from the kinetic theory.

COMPARISON OF CALCULATED WITH EXPERIMENTAL DATA

To offer a severe test of Eq. (11) a number of binary systems exhibiting highly irregular and dissimilar viscosity-concentration curves were selected from the literature. These data are plotted and compared with the calculated curves in Fig. 1: Properties of the pure components and literature sources for these systems are given in Table I. It is believed that the close correspondence between these calculated and observed data indicates the ability of Eq. (11) to reproduce gas viscosity behavior with sufficient precision for most purposes. The average deviation between calculation and experiment for the data shown in Fig. 1, excluding terminal points, is 0.97 percent. Prediction of the hydrogen-argon curve is the least satisfactory, and by excluding these data the average deviation between calculated and experimental points is reduced to 0.49 percent.

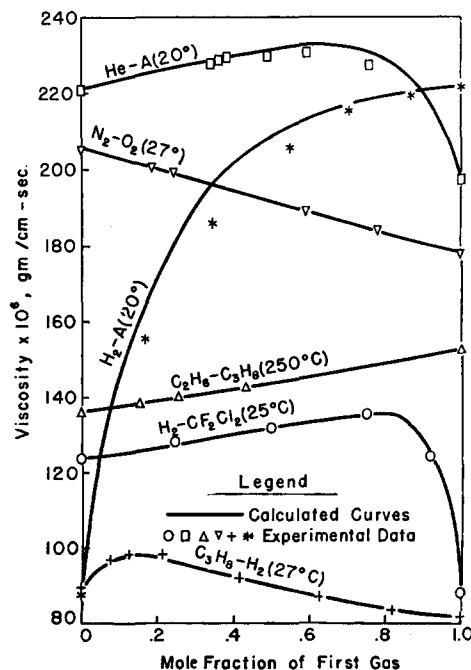


FIG. 1. Comparison of calculated and experimental viscosities for some binary mixtures.

TABLE III. Calculated and experimental data for some industrial gases.*

Components Component viscosity ^b	CO ₂	O ₂	CO	H ₂	CH ₄	N ₂	Higher hydro- carbons ^a	Viscosity × 10 ⁸ g/cm-sec.	
	147.2	201.9	174.9	87.55	108.7	178.1	90.9	μ_{calc}	μ_{exp}
Volume percent of components	6.2	10.7				83.1		178.1	179.3
Volume percent of components	10.4		28.5	1.6		59.5		172.9	173.8
Volume percent of components	10.6		29.8	3.9	0.3	55.4		172.4	174.3
Volume percent of components	2.5	0.8	14.9	53.0	18.1	9.1	1.6	142.7	135.5
Volume percent of components	2.2	1.0	4.0	52.3	29.9	9.4	1.2	130.7	130.7
Volume percent of components	4.8	0.3	26.4	17.2	2.6	48.2	0.5	171.4	171.4
Volume percent of components	3.5	0.3	27.3	14.4	3.7	50.0	0.8	171.1	171.2

* Data of F. Herning and L. Zipperer, Gas-u Wasserfach **79**, 49-54, 69-73 (1936), all at 293°K, atmospheric pressure.^a Assumed to be ethane.^b Based on data summarized by Hirschfelder, Bird, and Spotz (see reference 2).

GENERAL EQUATION FOR MULTI-COMPONENT SYSTEMS

Equation (2) may be written in a general form analogous to (3) and combined with equations such as (9) in the same manner as for the binary case to give a general viscosity equation for a mixture of any number of components. Thus:

$$\mu_m = \frac{\mu_1}{1 + (x_2/x_1)\phi_{12} + (x_3/x_1)\phi_{13} + \dots} + \frac{\mu_2}{1 + (x_1/x_2)\phi_{21} + (x_3/x_2)\phi_{23} + \dots} + \dots \quad (12)$$

Equation (12) may be written in the general form:

$$\mu_m = \sum_{i=1}^n \frac{\mu_i}{1 + \sum_{\substack{j=1 \\ j \neq i}}^n x_j \phi_{ij}} \quad (13)$$

where ϕ_{ij} is given by the equation:

$$\phi_{ij} = \frac{[1 + (\mu_i/\mu_j)^{\frac{1}{2}}(M_j/M_i)^{\frac{1}{2}}]^2}{(4/\sqrt{2})[1 + (M_i/M_j)^{\frac{1}{2}}]} \dots \quad (14)$$

Table II compares experimental data with those calculated by Eqs. (13) and (14), for several mixtures of three and four components. Pure component viscosities and calculated values of ϕ_{ij} for these systems are given

in Table I. Table III shows the results obtained for some industrial gases of three to seven components. The average deviation between calculated and experimental data for these thirteen mixtures is 1.9 percent. It is believed that these results demonstrate satisfactory application of the general equation to multicomponent systems.

NOMENCLATURE

D —diffusion coefficient, cm²/sec.; D_{11} , D_{22} , etc. for self-diffusion, D_{12} for a binary mixture.

k —Boltzmann's constant.

m —mass of a single molecule.

M —molecular weight.

n —number of molecules per unit volume in the portion of gas under consideration.

S —collision cross section for diffusion; S_{11} , S_{22} , etc. for pure components, S_{12} for dissimilar molecules.

t —temperature, °C.

T —temperature, °K.

x —mole fraction of a component in the mixture.

β —dimensionless constant.

λ —correction factor for deviation from inverse-fifth-power behavior.

μ —viscosity, g/cm-sec.; μ_1 , μ_2 , etc. refer to the pure components at the temperature and pressure of the mixture, μ_m is the viscosity of the mixture.

ϕ —dimensionless constant defined by Eq. (14).

ρ —density, g/cu.cm; ρ_1 , ρ_2 , etc. refer to the pure components at the temperature and total pressure of the mixture.

σ —molecular diameter.

Subscripts

i , 2 , i , j , n —refer to the pure components of the mixture.

m —refer to the mixture.